

ICB-2142
Examination –April/May- 2023
B.Tech. IV Sem :CSE-ICB
Analysis & Design of Algorithm

Time : 3 Hrs

Max. Marks : 70

Min. Marks : 22

Note: Total number of questions are 05. All Questions are compulsory. Each Question has 4 parts (a, b, c, d). Part a, b & c are compulsory while Part d has internal Choice. Assume missing data, if any.

Word limit be observed as follows:

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Part c – Max 100 words and Part d – Max 400 words.

Word limit NOT to be followed for diagram, numerical, derivation.

- Q.1 (a) Is $2^{n+1} = O(2^n)$ and Is $2^{2n} = O(2^n)$. Justify. 02
- (b) What is the time and space complexity of Heap sort? 02
- (c) Solve $T(n) = 2T(n/2) + 2$ if $n > 2$ and $T(n) = 1$ if $n = 2$. 03
- (d) Write the quick sort algorithm and derive its average case time complexity. 07

OR

Write down the Strassen's matrix multiplication algorithm. Derive its worst-case complexity. 07

- Q.2 (a) State the general principle of greedy method. 02
- (b) What is Huffman Coding? 02
- (c) Explain the characteristics of problems that can be solved using dynamic programming. 03
- (d) Find an optimal paranthesization of a matrix-chain product whose sequence of dimension is $4 \times 10, 10 \times 3, 3 \times 12, 12 \times 20, 20 \times 7$. 07

OR

Find an optimal solution to the knapsack instance $n = 7$ objects and the capacity of knapsack $m = 15$. The profits and weights of the objects are $(P_1, P_2, P_3, P_4, P_5, P_6, P_7) = (10, 5, 15, 7, 6, 18, 3)$ $(W_1, W_2, W_3, W_4, W_5, W_6, W_7) = (2, 3, 5, 7, 1, 4, 1)$ 07

- Q.3 (a) Define transitive closure of the directed graph. 02
- (b) Write the applications of BFS and DFS. 02
- (c) Define spanning tree of a graph. Write the total number of spanning trees possible for a complete graph with 6 vertices. 03
- (d) Write down DFS algorithm and analyze the time complexity. What is classification of edges that can be encountered during DFS operation and how is it classified? 07

OR

Write Dijkstra's single source of shortest path algorithm. Analyze its complexity. 07

16 14 10 18 15 + 3 + 2.5
 18 9 6 + 10 18 15 + 3 + 2.5
 18 9 6 + 10 18 15 + 3 + 2.5
 18 9 6 + 10 18 15 + 3 + 2.5

S.S

CSE-ICB-2142

- Q.4 (a) Define Chromatic number. 02
(b) Differentiate between the bounded and unbounded knapsack problem. 02
(c) Write about the constraints and criterion function used in backtracking. 03
(d) What is a Hamiltonian Cycle? Explain how to find Hamiltonian path and cycle using backtracking algorithm. 07

OR

How the travelling salesman problem can be solved using branch and bound. 07

- Q.5 (a) How is lower bound found by problem reduction? 02
(b) What are tractable and non-tractable problems? 02
(c) Depict the proof which says that a problem 'X' is no harder or no easier than problem 'Y' 03
(d) Discuss the approximation algorithm for NP-hard problem. 07

OR

What is class NP? Discuss about any three problems for which no polynomial time algorithm has been found. 07

NP

P

$n=2$

$$T(n) = 2T(n/2) + 2$$

$$= 2 + 2$$

$$T(2) = 4$$

r

CSE-1842
Examination - May-June - 2022
B.Tech. IV Sem : Computer Science & Engineering
Analysis & Design of Algorithm

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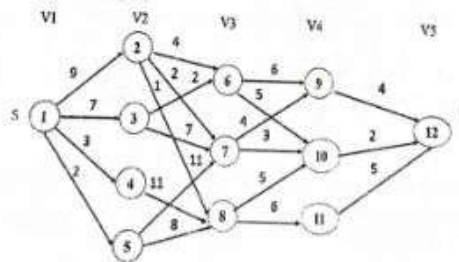
Min. Marks: 22

- Q.1 (a) Explain Master's Theorem for solving recurrence relation? 02
(b) Solve the recurrence relation $T(n) = 7T(n-1)$. 02
(c) What modification required in Linear search algorithm if you know that the given list is sorted. 03
(d) Discuss the criteria for analysis of algorithm and explain the importance of asymptotic order in analysis of algorithm. 07

OR

Write Strassen's algorithm of matrices multiplication and prove that it does $6n^{2.81} - 6n^2$ Addition Subtraction operations on matrix entries where n is power of 2. 07

- Q.2 (a) Explain the principle of optimality? 02
(b) Write the optimality function using dynamic programming for Longest common subsequence problem. 02
(c) Differentiate between Dynamic Programming and Divide and Conquer Technique. 03
(d) Find minimum path cost between vertex s and t for following multistage graph using dynamic programming. 07

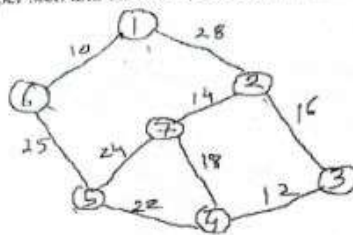


OR

How many bits may be required for encoding the message 'mississippi' using Huffman Coding Technique? 07

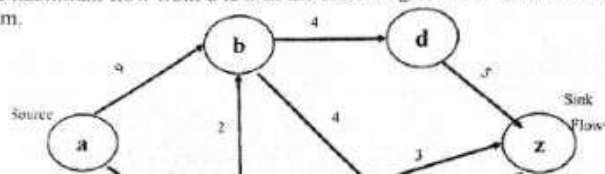
- Q.3 (a) Explain Union Find Data Structure with suitable example. 02
(b) A man wants to go different places in the world. He has listed them down all. But there are some places where he wants to visit before some other places. What application of graph can be used to determine that? 02

- (c) Differentiate between Depth First Search and Breadth First Search Graph Traversal techniques. 03
(d) Find out the minimum spanning tree for given graph using prim's algorithm. Solve the problem with proper step and also write the cost of minimum spanning tree. 07

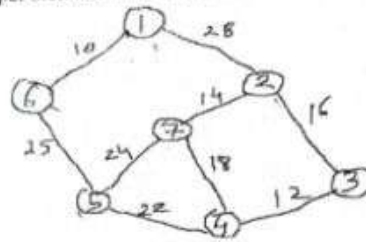


OR

Find the maximum flow from a to z in the following network graph using Network flow algorithm. 07

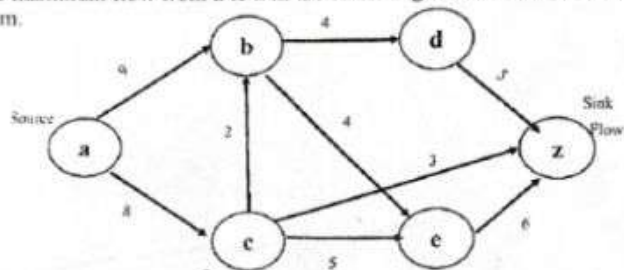


- (c) Differentiate between Depth First Search and Breadth First Search Graph Traversal techniques. 03
- (d) Find out the minimum spanning tree for given graph using prim's algorithm. Solve the problem with proper step and also write the cost of minimum spanning tree. 07



OR

Find the maximum flow from a to z in the following network graph using Network flow algorithm. 07



- Q.4 (a) Differentiate between explicit constraints and implicit constraints.
- (b) Briefly explain about least cost branch and bound method.
- (c) Solve the 4-queen problem using back tracking.
- (d) Solve the 15-puzzle problem using least cost branch bound method.

1	3	4	15
2		5	12
7	6	11	14
8	9	10	13



1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	

OR

Draw the portion of state space tree generated by Least Cost Branch and Bound for the following knapsack instance

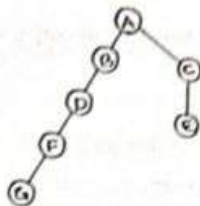
$N=4$, $P=(10, 10, 12, 18)$, $W=(2, 4, 6, 9)$ and $m=15$.

- Q.5 (a) What do you understand by NP-hard and NP-complete problem?
- (b) Write in brief About non-deterministic algorithm.
- (c) How 3-SAT Problem is help to solve NP-Hard Problem.
- (d) Design Horner's Algorithm to evaluate a polynomial. Give the time complexity of Your algorithm. Find the factors of the equation: $X^7+2X^6+6X^5+3X^4+7X^3+5X+4$

OR

Show that Hamiltonian Cycle Problem is NP-Complete.

- (d) Show In order, Preorder and Post order for following tree: 07



OR

Explain DFS and BFS with the help of an example. 07

(4)

Total Printed Pages: 04

Roll No.

CS/IT-1741

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- Q.1 (a) What is the purpose of Strassen's matrix multiplication? 02
- (b) What is the role of asymptotic notations in the study of algorithm? 02
- (c) Explain divide and conquer method and write an algorithm for in control abstraction. 03
- (d) Apply quick sort algorithm to sort the following list: 07
 37, 20, 18, 45, 98, 32, 10, 5

OR

Use binary search algorithm to search 26 from the given list: 07

2, 10, 48, 26, 17, 15, 29, 32, 42, 98, 31

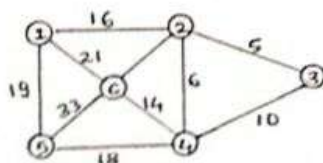
(1)

- Q.2 (a) What is greedy techniques? 02
- (b) Write down the basic difference between Prim's Algorithm and Kruskal's algorithm. 02
- (c) Solve the following job sequencing problem: 03
 $N = 5$, $(P_1, P_2, \dots, P_5) = (20, 15, 10, 5, 1)$ and $(d_1, \dots, d_5) = (2, 2, 1, 3, 3)$

- (d) Find the optimal binary merge tree (Pattern) for ten files whose length are 28, 32, 12, 5, 84, 53, 91, 35, 3 and 11. Also find in weighted external path length. 07

OR

Find shortest path from vertex ① to vertex ③ in the following graph using Dijkstra's algorithm. 07



- Q.3 (a) Write down the characteristics of dynamic programming. 02
- (b) Write algorithm for optimal solution of the knapsack problem using dynamic programming technique. 02
- (c) Explain how a reliability design can be obtained using dynamic programming. 03

(2)

- (d) What is multistage graph problem? Discuss in solution based on dynamic programming approach. Give suitable algorithm and find in computing time. 07

OR

Explain Floyd – Warshall algorithm with the help of an example. 07

- Q.4 (a) Write a Pseudo algorithm for graph coloring problem. 02
- (b) Explain the term lower bound with suitable example. 02
- (c) Give a backtracking algorithm for finding Hamiltonian cycles in a graph. 03

- (d) Solve the knapsack problem with $n = 3$, $m = 20$, $(P_1 = 25, P_2 = 24, P_3 = 15)$ and $(w_1, w_2, w_3) = (18, 15, 10)$ 07

OR

Solve the Traveling salesman problem using least cost branch and bound whose cost matrix is: 07

∞	7	3	12	8
3	∞	6	4	9
5	8	∞	6	18
9	3	5	∞	11
18	4	9	8	∞

- Q.5 (a) What are binary search trees? 02
- (b) Explain 2 – 3 trees with help of suitable example. 02
- (c) What is P, NP, class problem? Explain the concept with suitable example. 03

(3)